

PROJECT REPORT

TEAM ID:

**TITLE: Comprehensive Analysis and Dietary Strategies with Tableau:
A College Food Choices Case Study**

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1. INTRODUCTION

1.1 Project Overview

As a college health and wellness administrator, my mission is to proactively support the nutritional well-being of students. However, our current understanding of student dietary behaviours is hindered by fragmented, static, or siloed data. Without timely, actionable insights, our ability to address nutritional deficiencies and trends in real-time is limited - leaving us in a largely reactive position.

1.2 Purpose

The primary purpose of this project is to develop an interactive, data-driven platform using Tableau that enables college wellness administrators to:

- Understand and track student eating behaviours in real-time.
- Identify nutritional deficiencies and food trends across the student population.
- Enhance collaboration between health services and dining operations.
- Support evidence-based wellness programs and menu improvements that promote healthier eating on campus.

2. IDEATION PHASE

2.1 Problem Statement

I am a college health and wellness administrator

I'm trying to understand and improve the dietary and nutritional habits of students

But I lack real-time, visual insights into eating behaviors, health trends, and deficiencies

Because the data is scattered, static, or underutilized for strategic wellness interventions

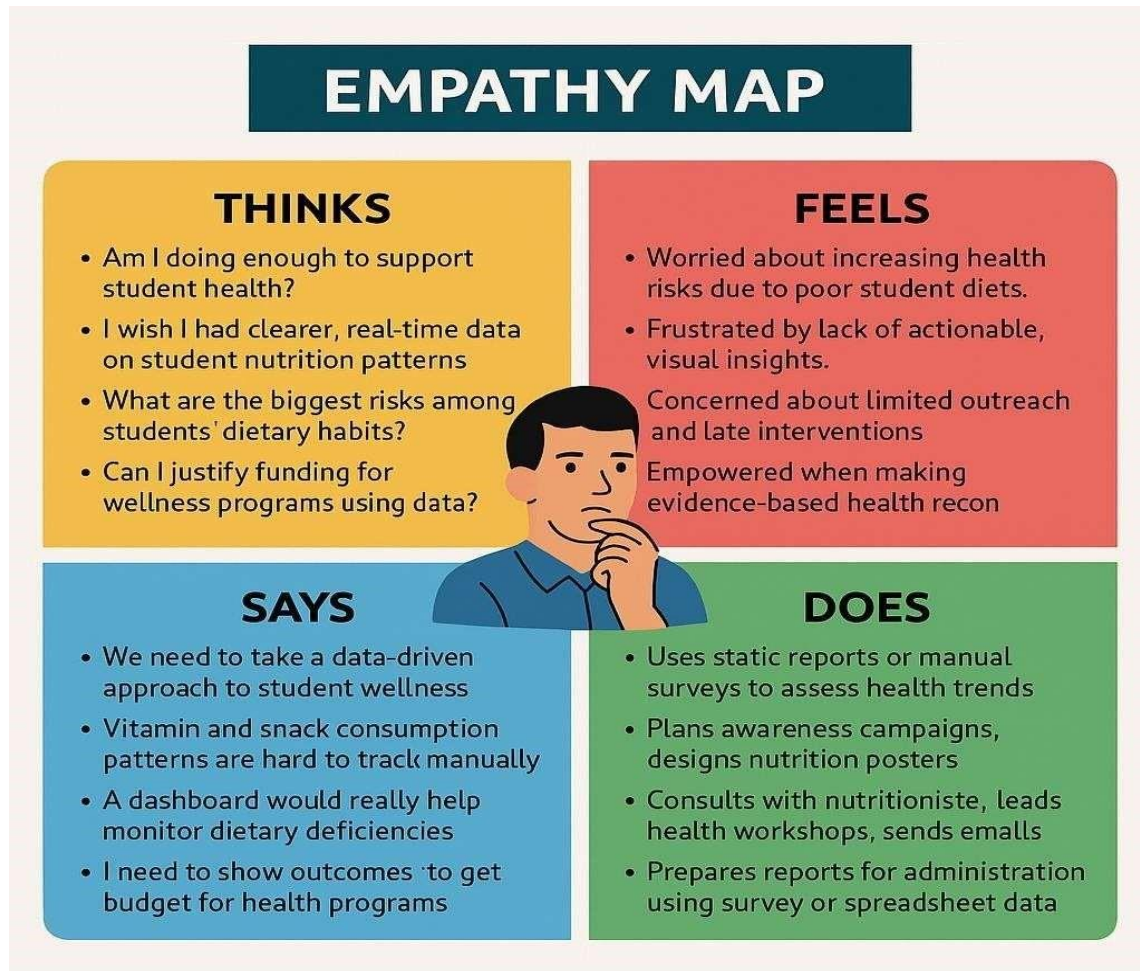
Which makes me feel reactive and under-equipped to support student health proactively.

PS-1: I am a student affairs officer who wants to monitor student nutrition habits to prevent health issues, but I don't have access to interactive tools that give real-time data on students' dietary trends.

PS-2: I am a university nutritionist trying to detect deficiencies in students' diets, but it's difficult to pinpoint problem areas without an integrated platform that shows snack overconsumption, vitamin neglect, and health perception metrics.

PS-3: I am a wellness strategist aiming to develop personalized health plans for students, but predictive analytics tools are not in place to proactively identify high-risk individuals and suggest tailored interventions.

2.2 Empathy Map



2.3 Brainstorming

Grouping Category	Ideas Generated
Nutritional Monitoring	Track fruit and vegetable intake trends by gender, age, and exercise frequency
Deficiency Detection	Identify vitamin intake gaps and high snack consumption patterns
Predictive Health Insights	Classify students into nutrition risk groups (High, Moderate, Low)
Visualization Tools	Use bar charts, scatter plots, heatmaps, and KPI cards in Tableau
Real-Time Alerts	Trigger alerts for low fruit/veggie intake or high snack calorie counts
Personalized Plans	Suggest meal plans and health guidance based on risk classification
Awareness Campaigns	Develop education initiatives based on dietary trends found in dashboards

Priority Level	Idea
High	✓ Build Tableau dashboards for real-time intake monitoring ✓ Classify students by risk using fruit and exercise data
Medium	✓ Visualize snack consumption vs. health perception ✓ Track vitamin intake by demographic
Low	✓ Link Tableau to live cafeteria menu data (future implementation) ✓ Integrate AI-based meal suggestions (post MVP phase)

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

CUSTOMER JOURNEY MAP

AWARENESS	CONSIDERATION	DECISION	ACTION	POST-ACTION
Customer Goal	Explore tools to visualize and track student nutrition patterns	Choose a data visualization platform to monitor and improve dietary trends	Implement visual dashboards and monitor student data	Evaluate impact, adjust strategies, and plan next interventions
Touchpoints	Research on tools e.g. Tableau, workshops health committee meetings	Tableau trials, case studies peer recommendations	Tableau dashboards, KPI alerts student reports	Follow-up dashboards wellness program feedback
Experience	Curious but overwhelmed by data complexity	Confident in Tableau's interactive dashboard capabilities	Empowered to make decisions based on trends and alerts	Reflects on dashboard insights and makes recommendations for next semester
Pain Points	No integrated system for diet tracking	Budget limitations, training needs for Tableau	Time consuming data cleaning interpreting visualizations	Difficulty measuring long-term change
Opportunities	Educate stakeholders about dietary data value	Show past success stories and predicted impact	Automate data updates schedule alerts	Expand dashboards with advanced analytics and student-level personalization

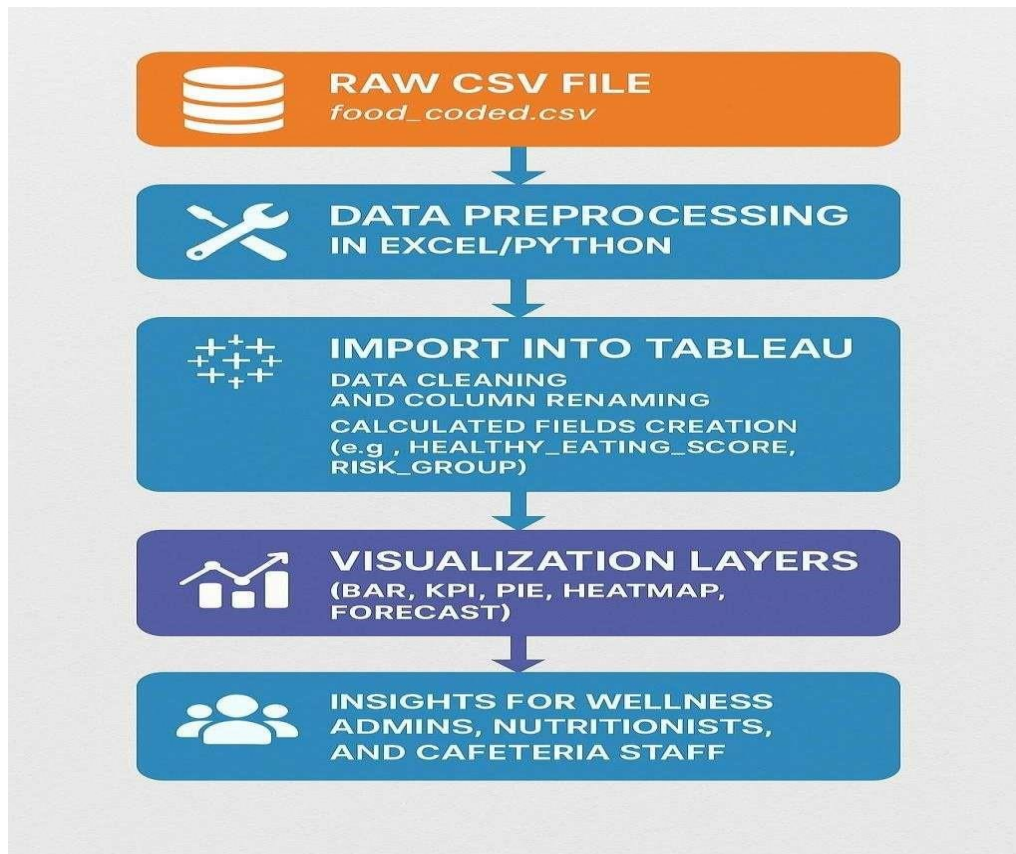
3.2 Solution Requirements Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Data Visualization	Generate Text table Generate Horizontal bar Chart Generate side-by-side bar Generate tree map Generate Dual line chart Generate packed bubble sort
FR-4	Data Filtering and Sorting	Filter by Data Range, Measure Names Sort by Calculations Filter by Gender

Non-Functional Requirements:

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should have a user-friendly and intuitive interface.
NFR-2	Security	Secure login and user data encryption should be implemented.
NFR-3	Reliability	The system should perform consistently under expected loads without failure.
NFR-4	Performance	Visualizations should load within 3 seconds on average.
NFR-5	Availability	The system should be accessible 24/7 with minimum downtime.
NFR-6	Scalability	Should be scalable to handle large datasets or more branches in the future.

3.3 Data Flow Diagram



3.4 Technology Stack

Data Layer

Component	Description
Dataset	food_coded.csv – structured CSV dataset capturing dietary and health info
Storage Format	Flat file (CSV) loaded locally for analysis
Tools Used	Excel (for cleaning & preprocessing if needed)

Data Processing Layer

Tool/Technology	Purpose
Excel	Initial cleaning or quick field review before importing to Tableau

Visualization & Analytics Layer

Tool/Technology	Purpose
Tableau	Main tool for interactive data visualization and dashboard creation
Tableau Features	KPI cards, Bar Charts, Pie Charts, Heatmaps, Highlight Tables, Forecasts
Calculated Fields	Used for: Consumption, Food Selection

User Interaction Layer

Feature	Role
Interactive Dashboards	Users can filter data by gender, risk, and exercise levels
Parameter Controls	Customize target values
Alerts & KPIs	Instant insight into nutrition deficiencies and trends
User Roles	Admin, Nutritionist, Cafeteria Staff – each interacts with filtered views

Security & Sharing

Feature	Notes
Tableau Public	Public dashboards (for non-sensitive data)
Tableau Server	Optional upgrade for secured, role-based access
Export Options	PDF reports, public link sharing, dashboard embedding

4. PROJECT DESIGN 4.1 Problem Solution Fit

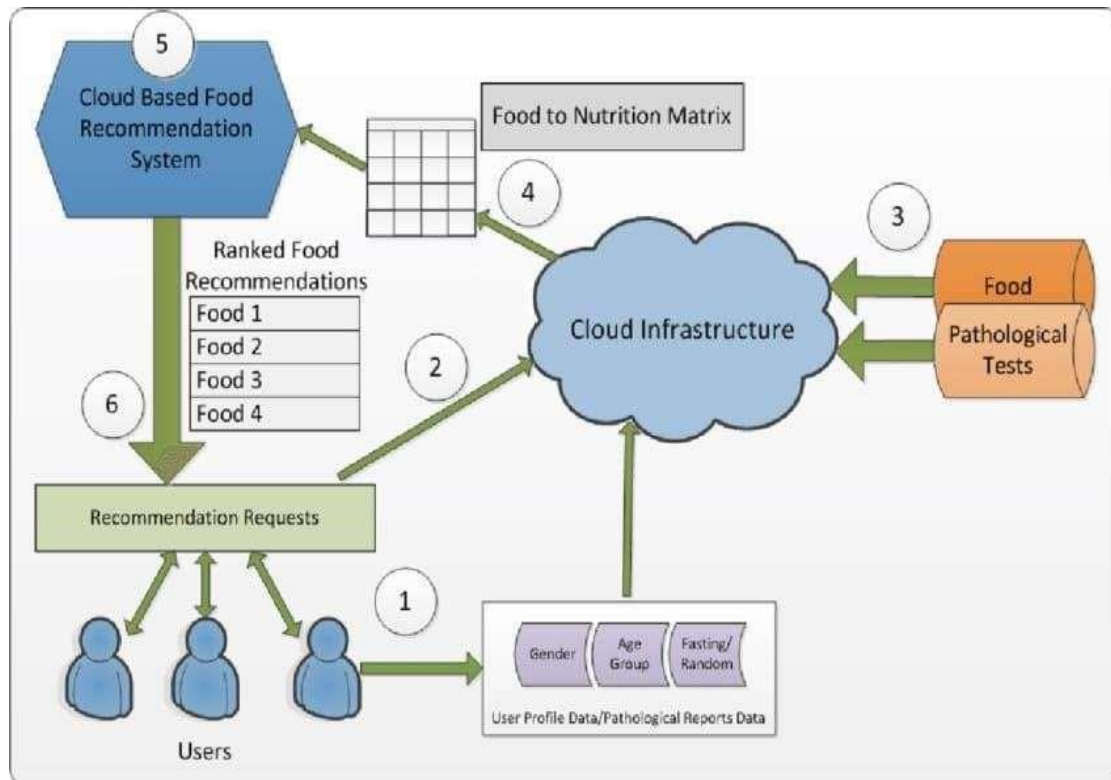
Define CS, fit into	1. CUSTOMER SEGMENT(S) CS College students and young working professionals aged 18–30 who are health-conscious but have poor diet routines due to lack of time, awareness, or planning tools.	6. CUSTOMER CC Low attention span or time Limited money for professional nutritionists No access to real-time food feedback Device limitations (some don't use laptops, only mobile) Overwhelmed by technical terms	5. AVAILABLE SOLUTIONS AS Mobile health apps (MyFitnessPal, HealthifyMe) YouTube diets / Instagram influencers:	Explore AS,
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Want to eat healthy but don't know how to start. Don't have time to plan meals or analyze food choices. Struggle to track nutrients or calories effectively. Feel overwhelmed by too much inconsistent advice online. Want to avoid lifestyle diseases but have no support system.	9. PROBLEM ROOT CAUSE RC Most customers lack structured, visual, and reliable food habit tracking tools tailored to their age/lifestyle. School and college curriculums often don't include practical nutrition education, leading to ignorance. They want to change, but the complexity of diet science makes it hard to start or stay consistent.	7. BEHAVIOUR BE Watch diet tips online Download free meal planners Try calorie counting apps Browse Google for quick-fix diets Sometimes buy health food impulsively	
Identify strong TR & EM	3. TRIGGERS TR Weight gain or health issues (e.g., acne, fatigue). Social media influence (seeing fitness transformations). Doctor advice to follow a better diet.	10. YOUR SOLUTION SL Analyzes historical food choice data Provides personalized food insights by gender, age, lifestyle Visualizes dietary patterns and gaps Suggests realistic meal changes	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE Follow diet influencers on Instagram or YouTube 8.2 OFFLINE Talk to gym instructors or peers Try following roommates' diets	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM Confused, guilty, overwhelmed, procrastinating Empowered, in control, health-aware, optimistic			

4.2 Proposed Solution

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Clearly describe the problem you aim to solve. Include relevant statistics or pain points to highlight the severity or urgency of the issue.
2.	Idea / Solution description	Outline your proposed solution. Focus on how it addresses the problem effectively and efficiently.
3.	Novelty / Uniqueness	What makes your solution different from existing ones? Highlight any innovative features, technologies, or processes.
4.	Social Impact / Customer Satisfaction	Explain how your solution improves lives, benefits communities, or enhances customer satisfaction.
5.	Business Model (Revenue Model)	How will the solution generate revenue? Include pricing strategies, partnerships, target customer segments, etc.

6.	Scalability of the Solution	Discuss how your solution can grow geographically or serve more users. Mention potential challenges and how they will be handled.
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4.3 Solution Architecture



5. PROJECT PLANNING AND SCHEDULING

5.1 Project Planning

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	2	High	3
Sprint-1	Confirmation	USN-2	As a user, I will receive confirmation email once I have registered for the application	1	High	3
Sprint-2	Registration	USN-3	As a user, I can register for the application through Facebook	2	Low	3
Sprint-1	Registration	USN-4	As a user, I can register for the application through Gmail	2	Medium	3

Sprint-1	Login	USN-5	As a user, I can log into the application by entering email & password	1	High	3
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Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	01 June 2026	08 June 2026	20	08 June 2026
Sprint-2	20	6 Days	09 June 2026	15 June 2026	20	15 June 2026
Sprint-3	20	6 Days	16 June 2026	22 June 2026	20	22 June 2026
Sprint-4	20	6 Days	23 June 2026	30 June 2026	20	30 June 2026

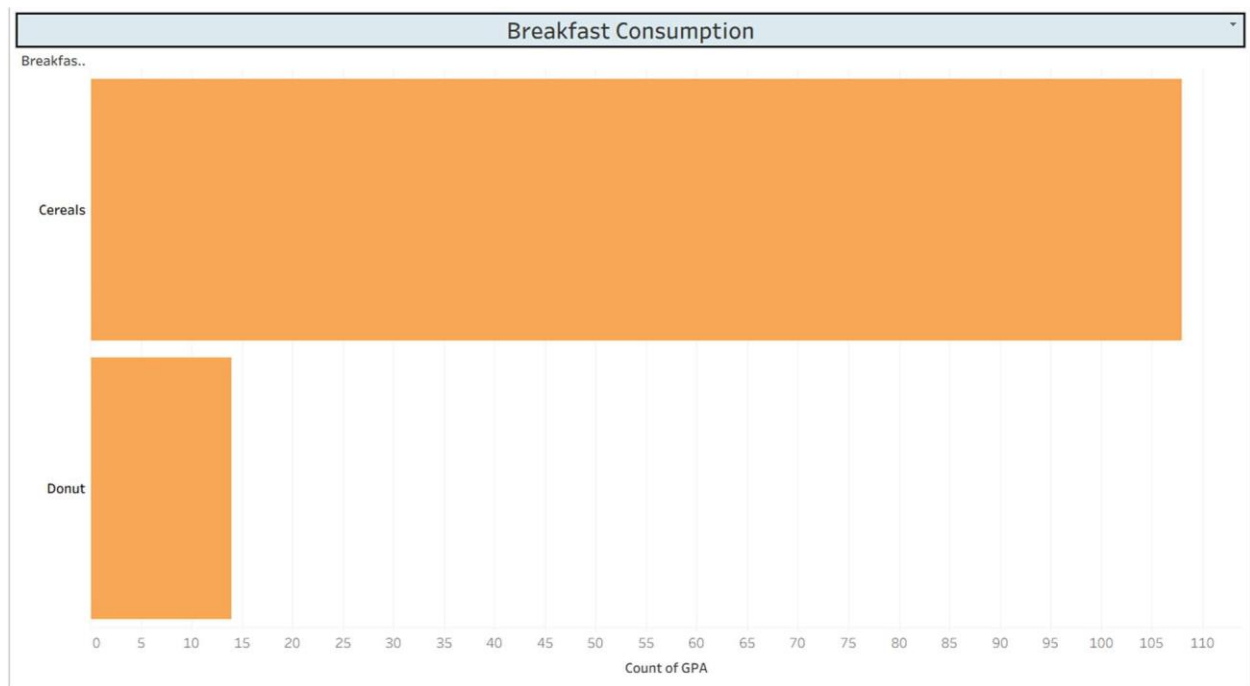
6. FUNCTIONAL AND PERFORMANCE TESTING

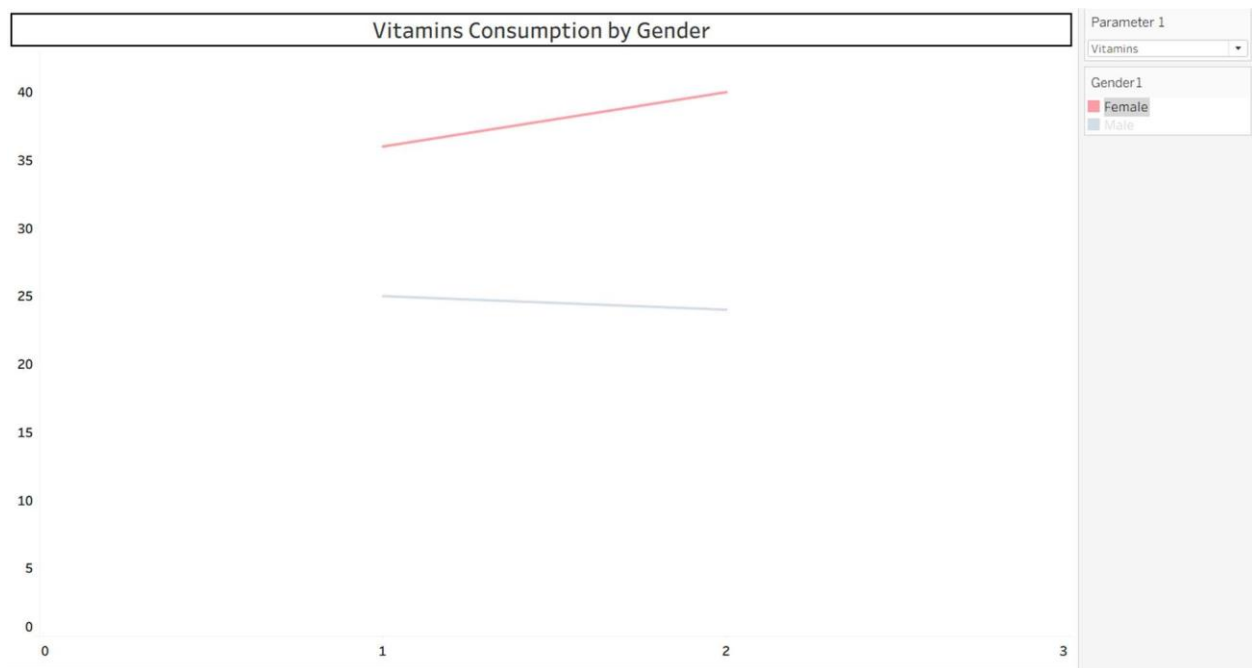
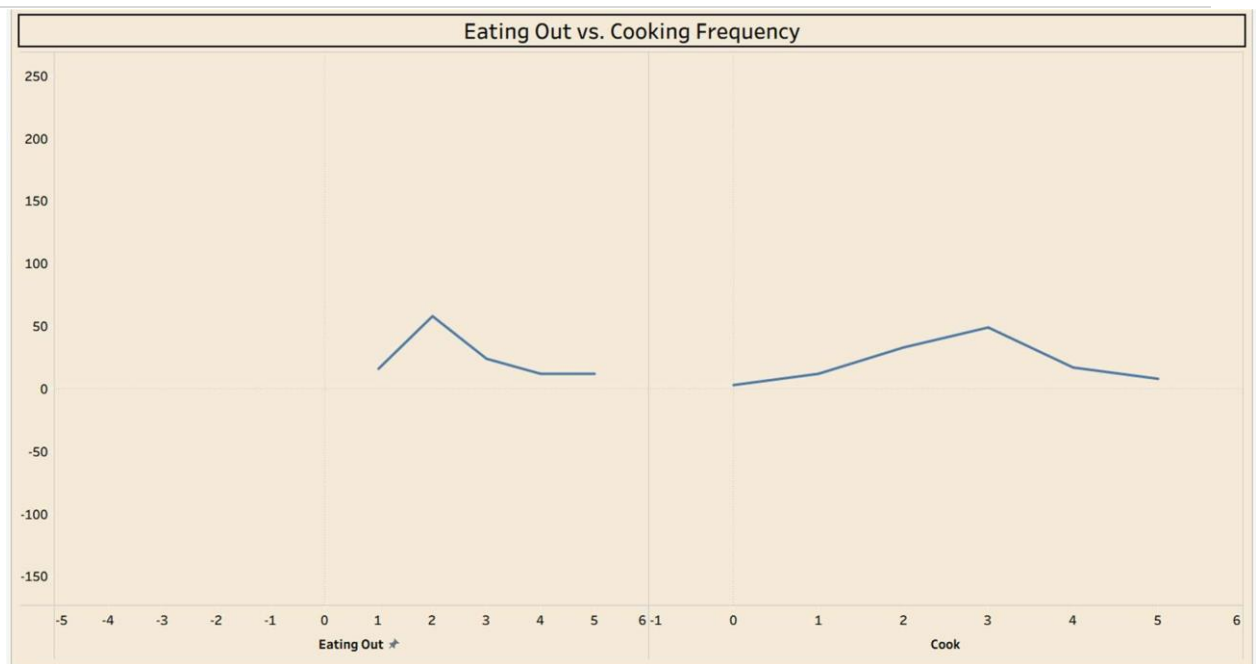
6.1 Performance Testing

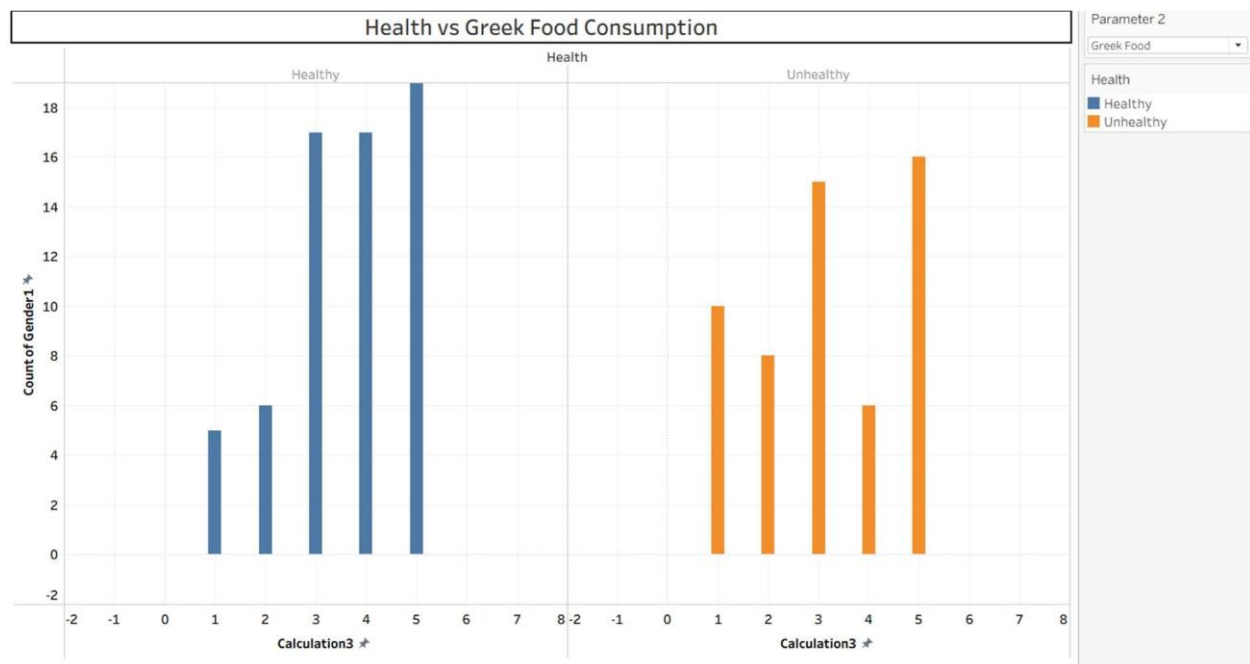
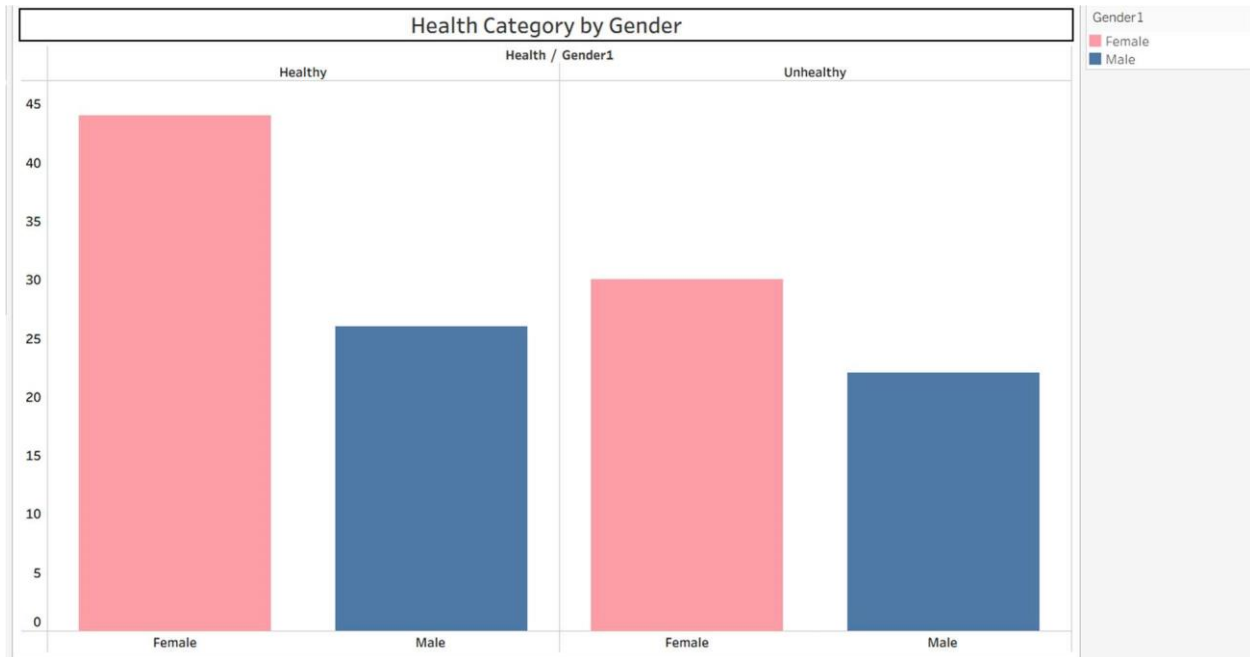
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	125 records
2.	Data Preprocessing	Removed null values
3.	Utilization of Filters	Date Range Filter Region Filter Product Category Filter
4.	Calculation fields Used	Usually numeric, date/time, or text depending on the formula and context.
5.	Dashboard design	No of Visualizations / Graphs -1
6	Story Design	No of Visualizations / Graphs -1

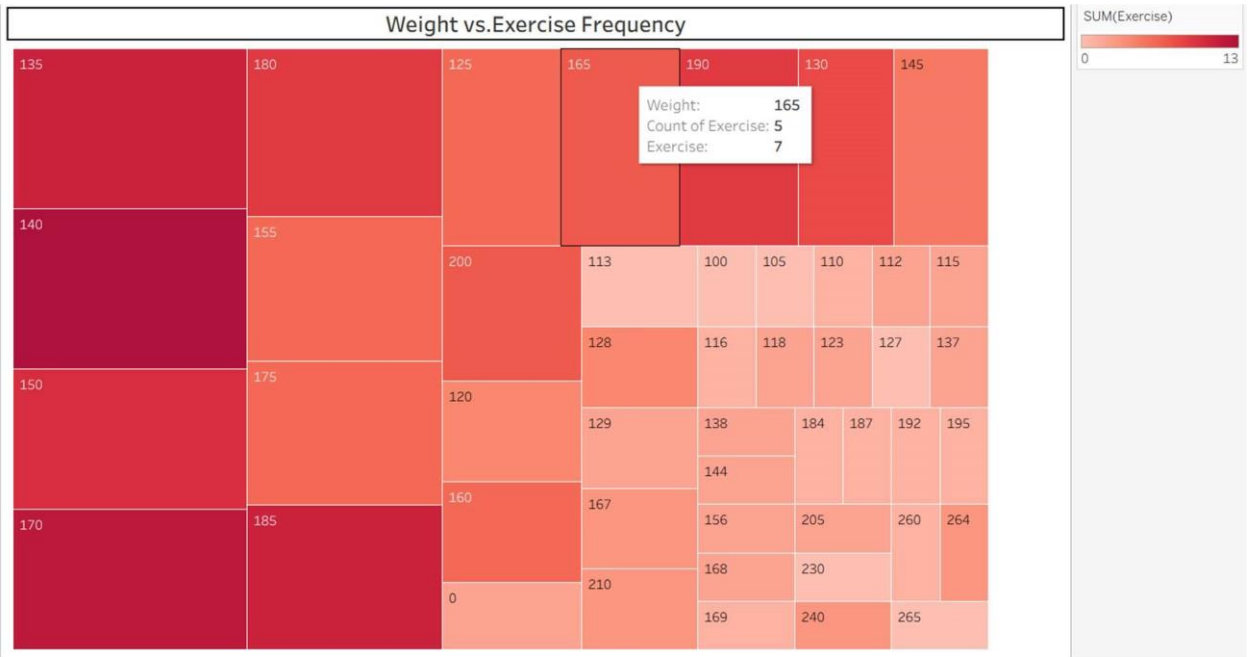
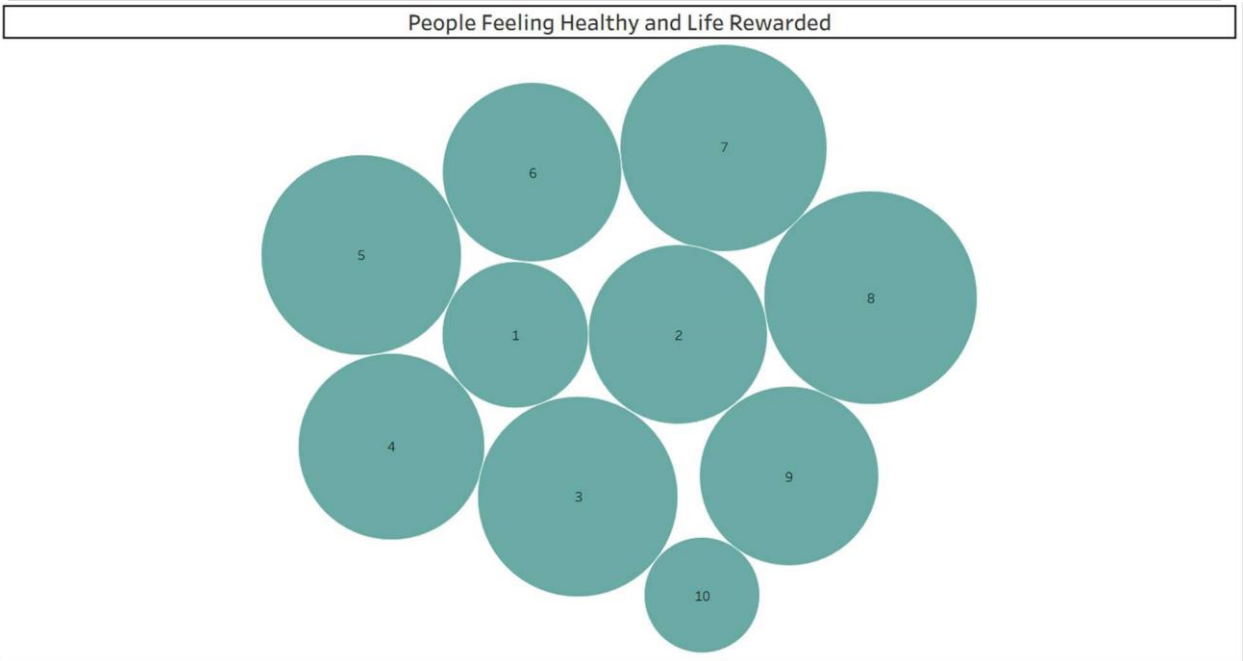
7. RESULTS 7.1 Output Screenshots

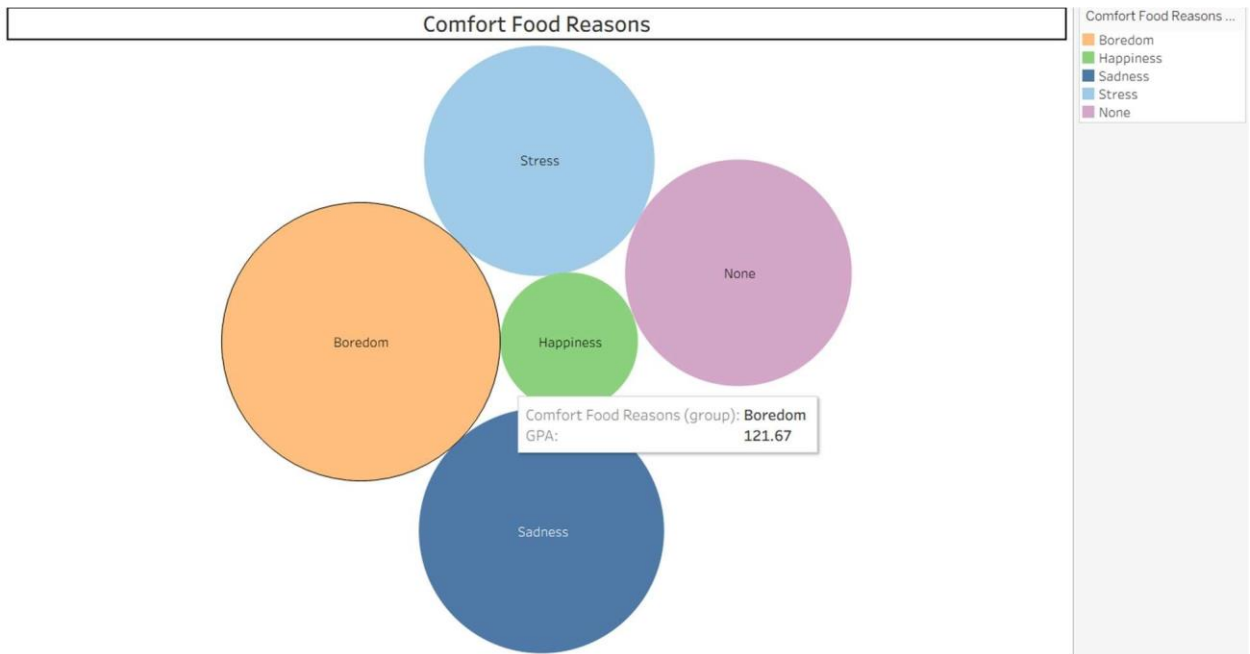
KPI Summary Card			
Count of GPA	Avg. GPA	Avg. Weight	Grade Level
122.0	3.4	156.7	297.0
Count of GPA	Avg. GPA	Avg. Weight	Grade Level



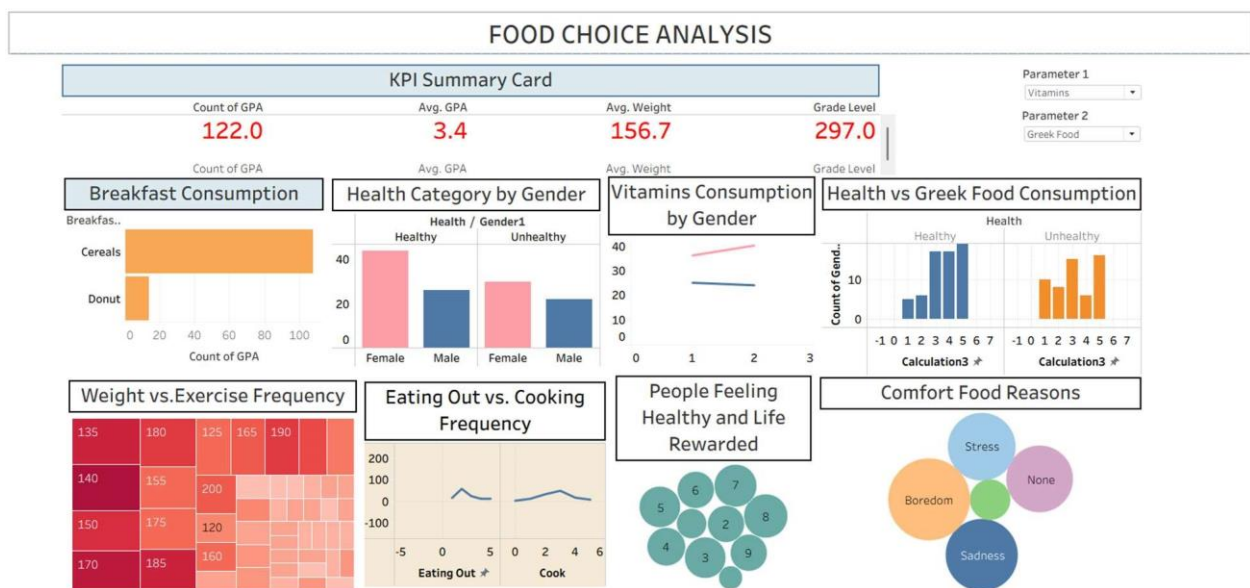








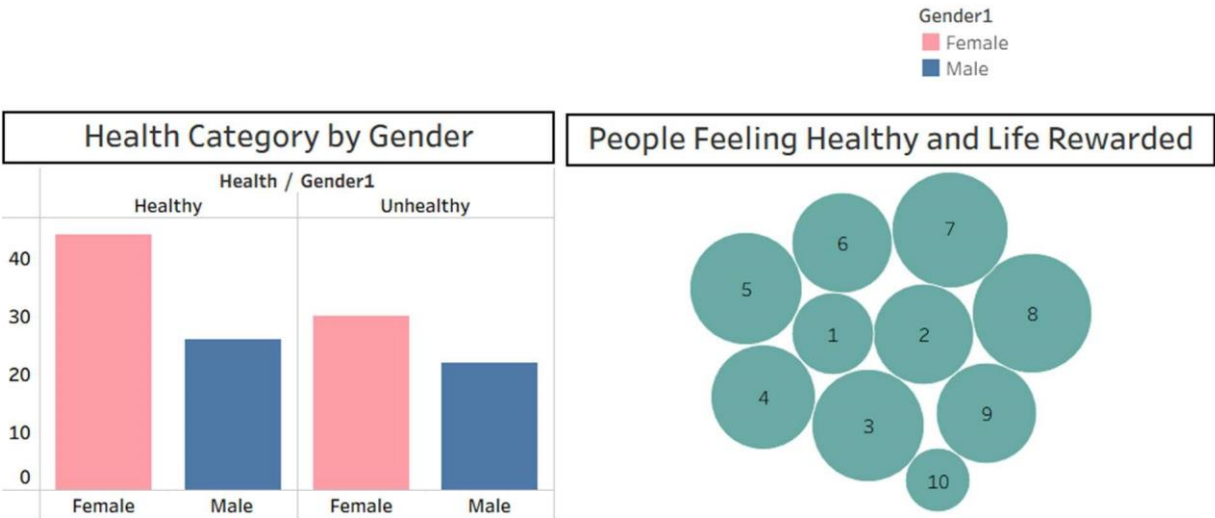
Dashboards:



Story:

Narrative & Analytical

How Gender Influences Healthy Classifications
and Impacts Life Satisfaction



8. ADVANTAGES AND DISADVANTAGES

ADVANTAGES:

- **Real-Time Insights:**
Tableau dashboards allow for up-to-date visualization of student dietary behaviours, helping administrators quickly spot trends and deficiencies.
- **Data-Driven Decision Making:**
Enables evidence-based strategies for improving menu planning, wellness programs, and targeted health interventions.
- **Centralized Data Integration:**
Combines multiple data sources (dining hall menus, nutrition data, student surveys, health records) into one accessible platform.
- **Improved Student Health Outcomes:**
Early identification of dietary deficiencies or poor eating habits can lead to timely support, better physical and mental health, and improved academic performance.
- **Enhanced Collaboration:**
Facilitates cooperation between health services, dining services, and student affairs through shared data insights.
- **Customizable Visual Dashboards:**
Dashboards can be tailored for different audiences—administrators, nutritionists, dining staff, or even students.
- **Cost-Effective Long-Term Planning:**
Helps avoid expensive health interventions later by focusing on prevention through nutrition.

DISADVANTAGES:

- **Data Collection and Quality Issues:**
Accurate and comprehensive data from multiple sources (e.g., dining systems, health records) may be hard to collect, inconsistent, or siloed.
- **Privacy and Ethics Concerns:**
Handling student health and dietary data requires strict compliance with privacy laws (e.g., FERPA, HIPAA) and ethical considerations.
- **Technical and Resource Limitations:**
Building and maintaining Tableau dashboards may require technical expertise, staff training, and ongoing IT support.
- **Limited Student Participation:**

Without consistent student input (e.g., food choices, survey responses), data may be incomplete or unrepresentative.

- **Resistance to Change:**
Dining services or administrative staff may resist adopting new data systems or changing long-standing menu practices.
- **Initial Time and Cost Investment:**
Upfront costs for data integration, software, and training may be significant, especially for smaller institutions.

9. CONCLUSION

The Comprehensive Analysis and Dietary Strategies with Tableau project represents a forward-thinking approach to improving student health and wellness through data-driven insights. By leveraging Tableau's powerful visualization capabilities, the project addresses key challenges such as fragmented data, lack of real-time analysis, and reactive health strategies. With integrated dashboards and nutritional analysis tools, college administrators, health professionals, and dining services can collaboratively identify dietary trends, detect nutritional gaps, and implement more effective, targeted interventions.

This initiative not only enhances operational efficiency but also places student well-being at the centre of campus life—fostering a healthier, more informed, and proactive campus community.

10. FUTURE SCOPE

1. **Predictive Health Analytics:**
 - Integrate machine learning tools with Tableau to forecast potential health issues related to diet (e.g., vitamin deficiencies, obesity risk).
2. **Mobile App Integration:**
 - Develop or integrate with mobile platforms to allow students to track their meals, receive personalized nutrition tips, and give feedback.
3. **Personalized Nutrition Plans:**
 - Use student dietary data to offer tailored meal recommendations based on individual health needs, allergies, or fitness goals.
4. **Expansion to Mental Health Insights:**
 - Correlate dietary trends with mental health patterns to support holistic student wellness initiatives.
5. **Collaboration with Academic Departments:**
 - Partner with nutrition, public health, or data science departments for student-led research, internships, or capstone projects.
6. **Benchmarking Across Institutions:**

- o Share anonymized insights across universities to compare trends, share best practices, and improve nationwide student health strategies.

7. Sustainability & Food Waste Monitoring:

Extend dashboards to analyse food waste and sustainability metrics, aligning dietary planning with environmental goals

11. APPENDIX

Dataset Link:

https://www.kaggle.com/datasets/borapajo/food-choices?select=food_coded.csv

GitHub Repository Link:

<https://github.com/Nagesh6969/Comprehensive-Analysis-and-Dietary-Strategies-with-Tableau>

Project Demo Link:

<https://drive.google.com/file/d/100BXQXAm000GXEPtFdZ8Od9-5IVx8bj9/view>

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